Amriteshkumar Yadav

Software Development Engineer

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Education

SRM Institute of Science and Technology — Chennai, India
B.Tech in Computer Science and Engineering (AI/ML Specialization), CGPA: 7.98/10 — *Graduated Jul 2026*

Technical Skills

Languages: Python, Go (Golang), C++, C, SQL, JavaScript, TypeScript, Bash
Backend & Architecture: REST APIs, Microservices, System Design, FastAPI, Node.js, React.js, Next.js
Databases & Messaging: PostgreSQL, Redis, Apache Kafka, DuckDB, Vector Databases (ChromaDB), SQLite
DevOps, Testing & Cloud: Docker, CI/CD, Prometheus, Grafana, OpenTelemetry, Unit & Integration Testing

Experience

Software Engineering Intern — Jul 2025 – Jan 2026
PopFix (now SuckSi) – University-Incubated Nutraceutical Startup — *Remote*

- Spearheaded **full-stack web development** (**React.js, Next.js**) and cloud setup as the **first technical hire**, configuring production domains, DNS routing, and automated order-processing workflows from scratch.
- Developed and deployed **Python automation** scripts for order tracking, vendor coordination, and logistics workflows, reducing manual processing time by **30%** and scaling operations to **100+ monthly orders**.
- Partnered with founders in go-to-market strategy meetings as the **sole technical voice**, translating business requirements into standardized fulfillment workflows that eliminated manual spreadsheet re-keying.

Projects

Fault-Tolerant ML Telemetry & Feature Streaming Pipeline — GitHub
Go (Chi, Goroutines), Python, Apache Kafka, PostgreSQL 16, Redis, Prometheus, Grafana, Docker

- Engineered a concurrent **Go HTTP ingestion service** using worker pools and Snappy producers, boosting throughput by **90% (288 RPS)**, cutting **p99 latency by 54.7%**, and reducing memory by **71.6% (24.3 MB)** vs. Python.
- Benchmarked **27 partition, consumer, and batch-size configurations** with a write-through Redis feature cache and PostgreSQL 16 storage, sustaining **7,117 events/sec (7,892 peak)** over 5+ minutes on a single host.
- Built a **Go dead-letter queue (DLQ) re-drive service** and JSON Schema validator with Prometheus metrics, routing **100% of 44,493 malformed events** with **zero data loss across 900K+ events** during container restarts.

Real-Time Warehouse Routing & Order-Picking Engine — GitHub
C++17, Python, FastAPI, Redis, pybind11, Docker, Locust, Pytest, Prometheus

- Engineered a **C++17 pathfinding engine** integrated with Python via pybind11, accelerating A* search across a 30,000-cell warehouse grid (**3,294 nodes expanded**) from **15.52 ms to 0.50 ms (31x speedup over Python)**.
- Built a multi-threaded **C++17 multi-stop route optimizer** (Held-Karp dynamic programming and 2-opt search) with cache-friendly memory layouts, cutting 10-stop order-picking latency from **73.57 ms to 1.30 ms (56.5x speedup)**.
- Designed a **FastAPI microservice** and async **Redis caching layer** with O(1) cache invalidation and Prometheus monitoring, reducing **median API latency by 43.8%** at **144 RPS** (**79.7% cache hit rate**; **0% errors**).

LLM Reliability Control Plane & RAG Evaluation Gateway — GitHub
C++, Python, FastAPI, Redis, OpenTelemetry (OTLP), DuckDB, ChromaDB, Llama 3.2 (Ollama), Docker

- Engineered an **OpenTelemetry (OTLP) trace ingestion gateway** in C++ and Python with CRC-32 validation and 64 MB WAL rotation, achieving **5 ms p99 ingest latency** (<**50 MB memory**) across **54 automated tests**.
- Implemented a **2-phase Zstd Parquet compaction** pipeline sustaining **50,000 spans/sec**, paired with a **Redis semantic cache** (90% cosine threshold) and **DuckDB analytical queries** completing in <**100 ms p99**.
- Built a **local RAG and hallucination guardrail pipeline** (**Llama 3.2 4-bit Q4_K_M**, ChromaDB, Pydantic) using Cross-Encoder verification, semantic similarity, and circuit-breaker fallbacks to block ungrounded outputs.

Publications

IEEE RMKMATE 2026: *"Workplace Safety Using Computer Vision: A Framework for Personal Protective Equipment Detection and Intrusion Monitoring"* (IEEE Xplore: 11518869 | GitHub) – Co-authored a real-time YOLO and OpenCV pipeline for PPE compliance and restricted-zone intrusion tracking.

Certifications & Achievements

Red Hat Certified System Administrator (RHCSA) | Oracle OCI Foundations Associate | Oracle Generative AI Professional (2025) | Quantum Quest Expo 2024 Finalist